

human

Where are houses in Exeter that were built before 1960?

host_agent

User wants to find houses in Exeter that were built before 1960. Provide the general steps to solve this geospatial query.

Tool: send_message

Arguments: {"target": "planning_agent", "task_description": "User wants to find houses in Exeter that were built before 1960. Provide the general steps to solve this geospatial query."}

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are houses, and we need to find houses in Exeter. The condition is that these houses must have been built before 1960.

output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Apply condition for houses built before 1960"]

Tool: send_message

Returned result: The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are houses, and we need to find houses in Exeter. The condition is that these houses must have been built before 1960.

output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Apply condition for houses built before 1960"]

Tool: send_message

Arguments: {"target": "named_area", "task_description": "Find the area polygon for Exeter."}

named_area

Find the area polygon for Exeter.

Tool: call_os_ngd

Arguments: {"filters": null, "bbox": null, "polygon_or_point": true, "street_address": null, "filename": "exeter_area_polygon"} Returned result: ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.,\n count of records fetched : 98.\n ',]

Tool: send_message

Arguments: {"target": "data_analysis_agent", "task_description": "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'.

Tool: generate_metadata_for_artifacts

Arguments: {"artifact_names": ["exeter_area_polygon"]}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where the name is 'Exeter' and description is 'City' or 'Settlement'
    filtered = gdf[((gdf['name1_text'].str.lower() == 'exeter') &
                    (gdf['description'].str.lower().isin(['city', 'settlement', 'part of settlement']))))
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    return [summary, 'filtered_exeter_area_polygon', 'GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and description.', filtered]
```

Filtered to 1 polygon for Exeter. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered by name and description.

Tool: send_message

Returned result: ["Filtered to 1 polygon for Exeter. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered by name and description.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and description.']", []]

named_area

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'.

Tool: send_message

Returned result: ["The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon'] and \n descriptions ['GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and description.']", []]

Tool: send_message

Arguments: {"target": "buildings", "task_description": "Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960."}

buildings

Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960.

Tool: call_os_ngd

Arguments: {"filters": null, "bbox": "filtered_exeter_area_polygon", "polygon_or_point": true, "street_address": false, "filename": "exeter_buildings_before_1960"} Returned result: ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['buildings_exeter_buildings_before_1960', 'buildingline_exeter_buildings_before_1960', 'buildingpart_exeter_buildings_before_1960'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None.", "A geopandas

dataframe containing building part data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None."],\n counts of records fetched are : [69169, 69342, 85480].\n ', [, ,]]

Tool: send_message

Arguments: {"target": "data_analysis_agent", "task_description": "Filter the artifact 'buildings_exeter_buildings_before_1960' to find houses built before 1960. Use the buildingage_period column to identify buildings constructed before 1999 and ensure buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Save the filtered results as 'filtered_houses_before_1960_exeter'. Provide a summary of the findings."}

data_analysis_agent

Filter the artifact 'buildings_exeter_buildings_before_1960' to find houses built before 1960. Use the buildingage_period column to identify buildings constructed before 1999 and ensure buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Save the filtered results as 'filtered_houses_before_1960_exeter'. Provide a summary of the findings.

Tool: generate_metadata_for_artifacts

Arguments: {"artifact_names": ["buildings_exeter_buildings_before_1960"]}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_houses_before_1960_exeter(data: list):
    import pandas as pd
    import geopandas as gpd
    # Assume data[0] is the buildings_exeter_buildings_before_1960 GeoDataFrame
    gdf = data[0]
    # Define periods before 1960
    periods_before_1960 = ['Pre-1837', '1837-1869', '1870-1918', '1919-1944', '1945-1959', 'Pre-1919']
    # Filter for houses built before 1960, residential, and single address
    filtered = gdf[
        (gdf['buildingage_period'].isin(periods_before_1960)) &
        (gdf['buildinguse_addresscount_residential'] > 0) &
        (gdf['buildinguse_addresscount_total'] == 1)
    ]
    summary = f"Filtered {len(filtered)} houses built before 1960 in Exeter with a single residential address."
```

```
artifact_name = 'filtered_houses_before_1960_exeter'
artifact_description = (
    "GeoDataFrame of houses in Exeter built before 1960, "
    "with exactly one residential address and no other address types. "
    "Filtered using 'buildingage_period', 'buildinguse_addresscount_residential', and 'buildinguse_addresscount_total'."
)
return [summary, artifact_name, artifact_description, filtered]
```

Filtered 19,786 houses built before 1960 in Exeter with a single residential address. The results are saved as 'filtered_houses_before_1960_exeter'. This artifact contains a GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address and no other address types, filtered using the columns 'buildingage_period', 'buildinguse_addresscount_residential', and 'buildinguse_addresscount_total'.

Tool: send_message

Returned result: ['Filtered 19,786 houses built before 1960 in Exeter with a single residential address. The results are saved as \'filtered_houses_before_1960_exeter\'. This artifact contains a GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address and no other address types, filtered using the columns \'buildingage_period\', \'buildinguse_addresscount_residential\', and \'buildinguse_addresscount_total\'. Additionally some data artifacts have been generated with names [\'filtered_houses_before_1960_exeter\'] and \n descriptions ["GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address and no other address types. Filtered using \'buildingage_period\', \'buildinguse_addresscount_residential\', and \'buildinguse_addresscount_total\'.'], []]

buildings

I found 19,786 houses in the 'filtered_exeter_area_polygon' that were built before 1960. The results are saved as 'filtered_houses_before_1960_exeter'. This dataset contains houses with exactly one residential address and no other address types.

Tool: send_message

Returned result: ['I found 19,786 houses in the \'filtered_exeter_area_polygon\' that were built before 1960. The results are saved as \'filtered_houses_before_1960_exeter\'. This dataset contains houses with exactly one residential address and no other address types. Additionally some data artifacts have been generated with names [\'filtered_houses_before_1960_exeter\'] and \n descriptions ["GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address and no other address types. Filtered using \'buildingage_period\', \'buildinguse_addresscount_residential\', and \'buildinguse_addresscount_total\'.'], []]

Tool: generate_metadata_for_all_artifacts

Arguments: {} Returned result: {'buildingline_exeter_buildings_before_1960': "A geopandas dataframe containing building line data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None.", 'buildingpart_exeter_buildings_before_1960': "A geopandas dataframe containing building part data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None.", 'buildings_exeter_buildings_before_1960': "A geopandas dataframe containing building data with filters and bbox applied for the query Find houses in the area defined by 'filtered_exeter_area_polygon' that were built before 1960. using filters None.", 'exeter_area_polygon': 'A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..', 'filtered_exeter_area_polygon': 'GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and description.', 'filtered_houses_before_1960_exeter': "GeoDataFrame of houses in Exeter built before 1960, with exactly one residential address and no other address types. Filtered using 'buildingage_period', 'buildinguse_addresscount_residential', and 'buildinguse_addresscount_total'."}

Tool: send_message

Arguments: {"target": "plotting_agent", "task_description": "Plot the houses from 'filtered_houses_before_1960_exeter' within the area of 'filtered_exeter_area_polygon'."}

plotting_agent

Plot the houses from 'filtered_houses_before_1960_exeter' within the area of 'filtered_exeter_area_polygon'.

Tool: generate_metadata_for_artifacts

Arguments: {"artifact_names": ["filtered_houses_before_1960_exeter", "filtered_exeter_area_polygon"]}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def plot_houses_within_exeter_area(data: list):
    import geopandas as gpd
    import folium
```

```

from shapely.geometry import mapping

# Unpack data
houses = data[0]
exeter_area = data[1]

# Ensure both are in the same CRS (EPSG:4326 for folium)
houses = houses.to_crs(epsg=4326)
exeter_area = exeter_area.to_crs(epsg=4326)

# Get the Exeter area polygon (assuming only one row)
exeter_polygon = exeter_area.iloc[0].geometry

# Filter houses within the Exeter area polygon
houses_within = houses[houses.within(exeter_polygon)]

# Create a folium map centered on the Exeter area
centroid = exeter_polygon.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Add Exeter area polygon to the map
folium.GeoJson(mapping(exeter_polygon),
                  name='Exeter Area',
                  style_function=lambda x: {'fillColor': 'blue', 'color': 'blue', 'weight': 2, 'fillOpacity': 0.1}).add_to(m)

# Add houses polygons to the map
folium.GeoJson(houses_within,
               name='Houses before 1960',
               style_function=lambda x: {'fillColor': 'orange', 'color': 'orange', 'weight': 1, 'fillOpacity': 0.5}).add_to(m)

# Add layer control
folium.LayerControl().add_to(m)

# Save map
map_filename = 'houses_within_exeter_area.html'
m.save(map_filename)

# Prepare summary
summary_rows = houses_within[['osid', 'description', 'buildingage_period', 'geometry_area_m2']].head(5)
summary = f"Plotted {len(houses_within)} houses built before 1960 within the Exeter area. First 5 entries:\n{summary_rows.to_string(index=False)}"

return [summary, map_filename, "Folium map showing houses built before 1960 within the Exeter area polygon. Houses are orange, Exeter area is blue.", map_f

```

Error: Object of type Timestamp is not JSON serializable

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
def plot_houses_within_exeter_area(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping
    import pandas as pd

    # Unpack data
    houses = data[0]
    exeter_area = data[1]

    # Ensure both are in the same CRS (EPSG:4326 for folium)
    houses = houses.to_crs(epsg=4326)
    exeter_area = exeter_area.to_crs(epsg=4326)

    # Get the Exeter area polygon (assuming only one row)
    exeter_polygon = exeter_area.iloc[0].geometry

    # Filter houses within the Exeter area polygon
    houses_within = houses[houses.within(exeter_polygon)]

    # Remove problematic columns for folium (non-serializable types)
    def clean_gdf(gdf):
        for col in gdf.columns:
            if pd.api.types.is_datetime64_any_dtype(gdf[col]) or pd.api.types.is_timedelta64_dtype(gdf[col]):
                gdf[col] = gdf[col].astype(str)
        return gdf
    houses_within = clean_gdf(houses_within)

    # Create a folium map centered on the Exeter area
    centroid = exeter_polygon.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Add Exeter area polygon to the map
    folium.GeoJson(mapping(exeter_polygon),
                    name='Exeter Area',
                    style_function=lambda x: {'fillColor': 'blue', 'color': 'blue', 'weight': 2, 'fillOpacity': 0.1}).add_to(m)

    # Add houses polygons to the map
    folium.GeoJson(houses_within,
                    name='Houses before 1960',
                    style_function=lambda x: {'fillColor': 'orange', 'color': 'orange', 'weight': 1, 'fillOpacity': 0.5}).add_to(m)

    # Add layer control
    folium.LayerControl().add_to(m)

    # Save map

```

```
map_filename = 'houses_within_exeter_area.html'
m.save(map_filename)

# Prepare summary
summary_rows = houses_within[['osid', 'description', 'buildingage_period', 'geometry_area_m2']].head(5)
summary = f"Plotted {len(houses_within)} houses built before 1960 within the Exeter area. First 5 entries:\n{summary_rows.to_string(index=False)}"

return [summary, map_filename, "Folium map showing houses built before 1960 within the Exeter area polygon. Houses are orange, Exeter area is blue.", map_f
```

Plotted 19784 houses built before 1960 within the Exeter area. First 5 entries: osid description buildingage_period geometry_area_m2
b9a49c80-fecf-4308-a98d-cf90ea4ea2e4 End-Of-Terrace House 1945-1959 99.338 9dfb4b41-f53d-44c0-8a7a-5f7884375677 Mid-Terrace House
1945-1959 78.565 aea058f2-6a39-4829-ae6b-85c6863a250e Mid-Terrace House 1945-1959 66.910 4d6aa883-1236-48ae-aca2-6e963070a974
End-Of-Terrace House 1945-1959 97.085 0847cf44-dbae-4f4a-b8ed-1e4016466f70 End-Of-Terrace House 1945-1959 83.259

plotting_agent

Plotted 19,784 houses built before 1960 within the Exeter area. First 5 entries: - b9a49c80-fecf-4308-a98d-cf90ea4ea2e4 | End-Of-Terrace House | 1945-1959 | 99.338 m² - 9dfb4b41-f53d-44c0-8a7a-5f7884375677 | Mid-Terrace House | 1945-1959 | 78.565 m² - aea058f2-6a39-4829-ae6b-85c6863a250e | Mid-Terrace House | 1945-1959 | 66.910 m² - 4d6aa883-1236-48ae-aca2-6e963070a974 | End-Of-Terrace House | 1945-1959 | 97.085 m² - 0847cf44-dbae-4f4a-b8ed-1e4016466f70 | End-Of-Terrace House | 1945-1959 | 83.259 m²

Artifact: houses_within_exeter_area.html

Description: Folium map showing houses built before 1960 within the Exeter area polygon. Houses are orange, Exeter area is blue. File: houses_within_exeter_area.html

host_agent

Here are the results for your query:

- There are 19,784 houses in Exeter that were built before 1960.
- These houses are distributed throughout the Exeter area.
- The houses are filtered to include only those with exactly one residential address and no other address types.

A map has been generated showing all these houses (in orange) within the Exeter area (in blue).

If you would like to view or download the map, let me know! If you need more details or want to refine the search (e.g., by neighborhood or house type), please specify.